Cartesian coordinates

$$c = E, p_x, p_y, p_z$$

Polar cooordinates

$$p = \eta, \phi, p_T, E$$

where

$$\begin{aligned}
\eta &= \operatorname{arcsinh}\left(\frac{p_z}{\sqrt{p_x^2 + p_y^2}}\right) = \operatorname{arcsinh}\left(\frac{p_z}{p_T}\right) = \ln\left[\frac{p_z}{p_T} + \sqrt{\left(\frac{p_z}{p_T}\right)^2 + 1}\right] \\
\phi &= \operatorname{arctan2}(p_y, p_x) \\
p_T &= \sqrt{p_x^2 + p_y^2} \\
E &= E
\end{aligned}$$

$$\begin{aligned}
&\begin{bmatrix}
0 & -\frac{p_x\,p_z}{\left(p_x^2+p_y^2\right)^{\frac{3}{2}}\sqrt{\frac{p_z^2}{p_x^2+p_y^2}+1}} & -\frac{p_y\,p_z}{\left(p_x^2+p_y^2\right)^{\frac{3}{2}}\sqrt{\frac{p_z^2}{p_x^2+p_y^2}+1}} & \frac{1}{\sqrt{p_x^2+p_y^2}\sqrt{\frac{p_z^2}{p_x^2+p_y^2}+1}} \\
0 & -\frac{p_y}{p_x^2+p_y^2} & \frac{p_x}{p_x^2+p_y^2} & 0 \\
0 & \frac{p_x}{\sqrt{p_x^2+p_y^2}} & \frac{p_y}{\sqrt{p_x^2+p_y^2}} & 0 \\
1 & 0 & 0 & 0
\end{bmatrix} \\
= &\begin{bmatrix}
0 & -\frac{p_x\,p_z}{p_T^3\sqrt{\frac{p_z^2}{p_T^2}+1}} & -\frac{p_y\,p_z}{p_T^3\sqrt{\frac{p_z^2}{p_T^2}+1}} & \frac{1}{p_T\sqrt{\frac{p_z^2}{p_T^2}+1}} \\
0 & -\frac{p_y}{p_T^2} & \frac{p_x}{p_T^2} & 0 \\
0 & \frac{p_x}{p_T} & \frac{p_y}{p_T} & 0 \\
1 & 0 & 0 & 0
\end{bmatrix}
\end{aligned}$$